

Block 1: Research Context

RAG grounds language-model answers in external documents, but legal QA requires a pipeline to satisfy two goals at once: retrieve enough relevant evidence to produce a correct answer, while keeping source-page citations precise and complete. This is challenging because legal PDFs may lack machine-readable text, evidence can appear at different textual scales, and questions may depend on either exact legal references or semantic similarity. Moreover, retrieving more evidence can improve coverage while also introducing irrelevant source pages. We therefore examine how design choices in document processing, retrieval, and attribution can improve answer quality without sacrificing citation quality.

Block 2: Prior Work & Study Focus

Retrieval-augmented generation (RAG) conditions language-model answers on retrieved documents [1]. Legal retrieval must balance precision and coverage: LegalBench-RAG targets minimal relevant passages [2], while regulatory QA studies show that retrieving all supporting evidence becomes harder when it spans multiple passages [3]. In an end-to-end legal benchmark, retrieval-model choice had a larger effect on answer correctness than generator choice [4].

Evaluation also extends beyond correctness. Legal QA research examines whether answers are supported by the supplied context [5], while ALCE evaluates answer correctness and citation quality separately [6]. Together, these studies motivate evaluating both the evidence available to the generator and the sources ultimately cited.

Our study complements these approaches by holding the generator fixed and examining a broader sequence of pipeline choices, from document processing to final page selection.

Block 3: Research question

With the answer generator fixed, which pipeline additions improve answer quality and page-level citation precision and recall, and where do these objectives conflict?

Block 4: Our Contribution

We build a Legal RAG pipeline and provide an empirical comparison of seven cumulative configurations on 30 documents and 100 questions. By adding

components sequentially while keeping the answer generator fixed, we measure changes in answer quality and page-level citation precision and recall across stages of document processing, retrieval, reranking, answer processing, and citation selection.

Block 5: Hypothesis

1. H1 — Retrieval

- a. Combining multiple text granularities, semantic and lexical retrieval, and relevance reranking will improve answer quality and evidence retrieval compared with a simple semantic-retrieval baseline.

2. H2 — Robustness to scanned documents

- a. Recovering text from scan-like legal documents will reduce the performance loss caused by the absence of a machine-readable text layer.

3. H3 — Citation attribution

- a. Selecting source pages after answer generation will produce a more focused citation set without substantially reducing the coverage of relevant evidence.

Block 6: Data and Methods

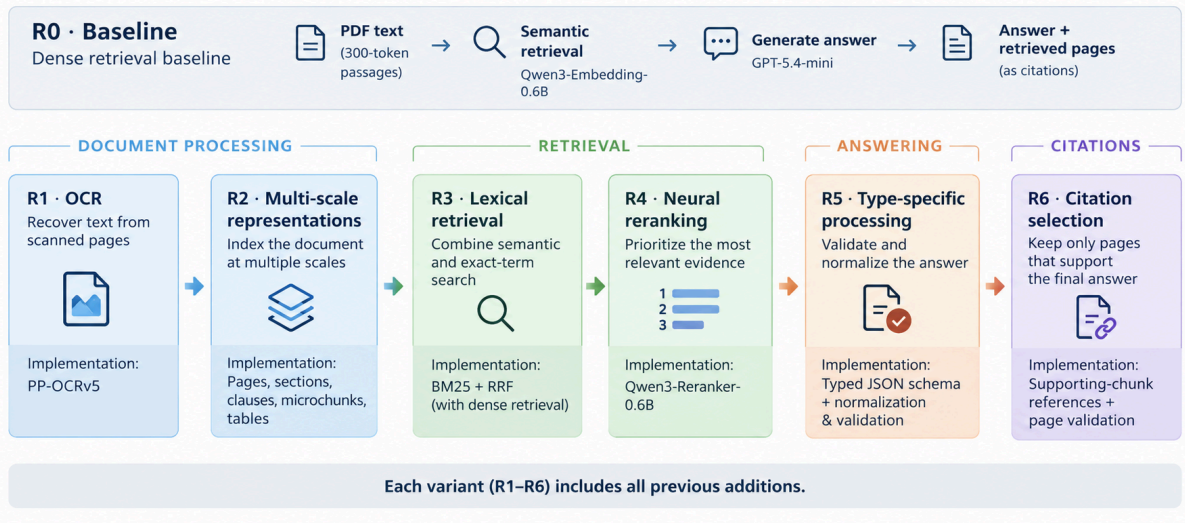
Data

30 DIFC legal PDFs (590 pages), 100 questions, including 70 structured-answer and 30 free-text questions. Reference source pages are available for 95 questions. For the scan-robustness test, 10 of the 30 documents (107 pages) were converted into synthetic scans: each page was rasterised as an image and its machine-readable text layer was removed.

Pipeline

02 METHODOLOGY

We start with a simple Legal RAG baseline and add one component at a time.



Starting from a simple baseline, we add one capability at a time. Each variant includes all previous additions.

R0 · Baseline

- PDF text → 300-token passages → semantic retrieval → GPT-5.4-mini → answer + retrieved pages
- *Dense retrieval: Qwen3-Embedding-0.6B*

DOCUMENT PROCESSING

- **R1 · Read scans**
Convert scanned pages into searchable text
PP-OCRv5
- **R2 · Represent evidence at multiple scales**
Pages · sections · clauses · microchunks · tables

→

RETRIEVAL

- **R3 · Find exact terms too**
Combine semantic and exact-word search
BM25 + RRF
- **R4 · Prioritize better evidence**
Move the most relevant passages to the top
Qwen3-Reranker

→

ANSWERING

- **R5 · Check the answer**

Normalize and validate the expected answer type

Typed schema

→

CITATIONS

- **R6 · Refine citations**

Keep only pages that support the final answer

Supporting-chunk references

Answer generation: A fixed GPT-5.4-mini configuration is used across variants, so changes in performance reflect modifications to the surrounding pipeline rather than a change of the generating LLM.

Block 7: Evaluation

A correct answer does not necessarily imply good citations. A system may produce the right answer while citing irrelevant pages, or it may cite only part of the evidence required to support the answer. We therefore evaluate answer quality and citation quality separately.

Answer quality

Is the generated answer correct?

$Q = 0.7 \times \text{structured score} + 0.3 \times \text{free-text score}$

Citation precision

Are the cited pages actually relevant?

Share of cited pages that support the answer.

Citation recall

Did the system cite all required evidence?

Share of annotated supporting pages that were cited.

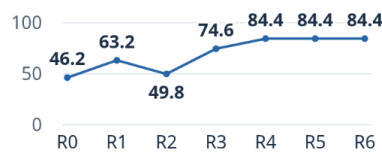
Block 8: - Results

Keep graphs from previous design

03 RESULTS

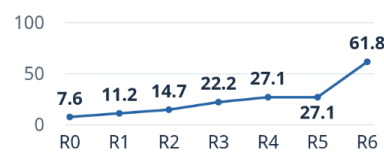
Answer quality Q, %

100 questions



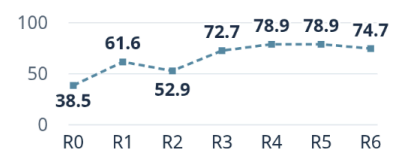
Citation precision, %

95 questions



Citation recall, %

95 questions



R0 to R6 on the mixed corpus

Sub-block 9: OCR recovers performance on scanned documents

- Original corpus · R0 — 65.4
- Mixed corpus (original + synthetic scans) · R0 — 46.2
- Mixed corpus + OCR · R1 — 63.2

Под ним:

Converting 10 documents into synthetic scans reduces answer quality from 65.4% to 46.2%. OCR restores most of this loss, reaching 63.2%.

Block 10: Hypothesis evaluation

Hypothesis

Result

H1 — Retrieval & representation

Partially supported. Multi-scale representations alone reduce Q, but adding lexical retrieval and reranking substantially improves answer quality and source-page recall.

H2 — Scan robustness

Supported. OCR restores most of the answer-quality loss caused by removing the machine-readable text layer.

H3 — Citation selection

Supported with a trade-off. Precision rises from **27.1% to 61.8%**, while recall decreases moderately from **78.9% to 74.7%**.

Block 11: Conclusion and takeaways

Main takeaway

Legal RAG should be designed as a multi-stage pipeline: OCR improves robustness to unreadable PDFs, hybrid retrieval and reranking improve answer

quality, and post-answer citation selection improves citation precision. However, citation selection introduces a precision–recall trade-off, so answer quality and citation quality should be optimized and evaluated separately.

Additional Takeaways:

1 · Ensure document robustness

OCR provides robustness when machine-readable text is unavailable.
It recovered most of the performance loss introduced by synthetic scans.

2 · Optimize retrieval

Hybrid retrieval and reranking produced the largest observed gains in answer quality. Additional representation complexity did not consistently improve performance.

3 · Optimize citations separately

Post-answer citation selection substantially increased precision, with a small reduction in recall.

Unexpected outcome: Multi-scale representations did not improve performance; Q decreased from 63.2% to 49.8% after R2.

Block 12: Limitations

one DIFC corpus and 100 questions; one run per condition with no significance testing; LLM-based evaluation of free-text answers.

Block 13: References

- 1 - Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., ... & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems*, 33, 9459-9474.
- 2 - Pipitone, Nicholas, and Ghita Hourir Alami. "Legalbench-rag: A benchmark for retrieval-augmented generation in the legal domain." *arXiv preprint arXiv:2408.10343* (2024).
- 3 - Gokhan, T., & Briscoe, T. (2025, November). Grounded Answers from Multi-Passage Regulations: Learning-to-Rank for Regulatory RAG. In *Proceedings of the Natural Legal Language Processing Workshop 2025* (pp. 135-146).

4 - Butler, A. R., & Butler, U. (2026). Legal RAG Bench: an end-to-end benchmark for legal RAG. *arXiv preprint arXiv:2603.01710*.

5 - Trautmann, D., Ostapuk, N., Grail, Q., Pol, A., Bonifazi, G., Gao, S., & Gajek, M. (2024, November). Measuring the groundedness of legal question-answering systems. In *Proceedings of the Natural Legal Language Processing Workshop 2024* (pp. 176-186).

6 - Gao, T., Yen, H., Yu, J., & Chen, D. (2023, December). Enabling large language models to generate text with citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing* (pp. 6465-6488).